

An attention-based balanced variational autoencoder method for credit card fraud detection

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ABSTRACT

Credit card fraud detection poses a significant role in both academia and industries. Conventionally, manual checking is time-consuming and not accurate. With the advent of artificial intelligence (AI), numerous machine-learning approaches have emerged. Most of them are proven effective and efficient, especially those based on attention mechanisms. However, the highly imbalanced distribution of normal and fraudulent transactions increases the hardness of further performance improvement. So far, there exist three issues in current contributions: 1) the existing oversampling methods did not fully consider the imbalanced distribution theoretically; 2) the current methods neglected the importance of features and did not mine helpful information from them; 3) the layers of current attention-based methods are too shallow to capture enough characteristics of data. To address the above issues, we propose a novel method: Balanced-Variational AutoEncoder-Attention (Bal-VAE-Attention), which can significantly boost the existing fraud detection results. In our approach, we design a novel oversampling method that considers the imbalanced data distribution. Moreover, we apply an automatic feature selection procedure. Also, we deploy a deep multi-head attention structure to depict the complex inner structure of fraudulent data. We implement abundant experiments on two open-source credit card fraud datasets. Through substantial experiments and ablation study, we prove our proposed method's effectiveness. To the best of our knowledge, it is superior to other state-of-the-art credit card fraud detection baselines.

1. Introduction

In modern society, people are seeking more secure approaches to obtain financial services. As one of the core elements of finance, credit card-related techniques have drawn the attention of both academia and industries. Credit card fraud is defined as the unauthorized consumption of funds within a transaction by credit or debit cards [1]. With the rising number of credit card owners, fraudulent transactions have emerged. According to the Nilson Report (December 2021), global credit card and debit card fraud gave rise to losses of 28.58 billion US dollars during 2020 [2]. These huge losses lead to the extreme importance of credit card fraud detection.

Conventionally, credit card fraud detection was conducted manually, with complex and time-consuming checking and justification. As depicted in Fig. 1, the credit card fraud detection process mainly includes three steps: checking card transactions, verification, and making final decisions. These procedures are optimized with the advent of machine learning and AI tools. AI-driven detection can be automatically conducted and costs less time in verification and decision.

In academia, a series of AI methods have shown extraordinary capabilities in dealing with frauds. In particular, models with attention mechanism perform extremely well because of the abilities of Transformer to tackle sequential data [3–7]. For instance, Valliammal Narayan et al. proposed a hybrid deep learning method HS-SAL-BiLSTM to detect credit card fraud and improved the results of traditional methods [4]. LSTM-Attention was proposed in [3] to consider the sequential essence of data and enhance performance with attention. Moreover, due to the highly imbalanced distribution of fraudulent and normal data, some oversampling methods were applied and outperformed those without oversampling and augmentation. For example, sparse autoencoder (SAE) and variational autoencoder (VAE) were applied to enhance the original feature information [8,9]. However, there are three major issues with current contributions.

First, the existing oversampling methods did not fully consider the imbalanced distribution theoretically. For instance, the vanilla VAE method adopts KL divergence as its main ingredient of loss function.

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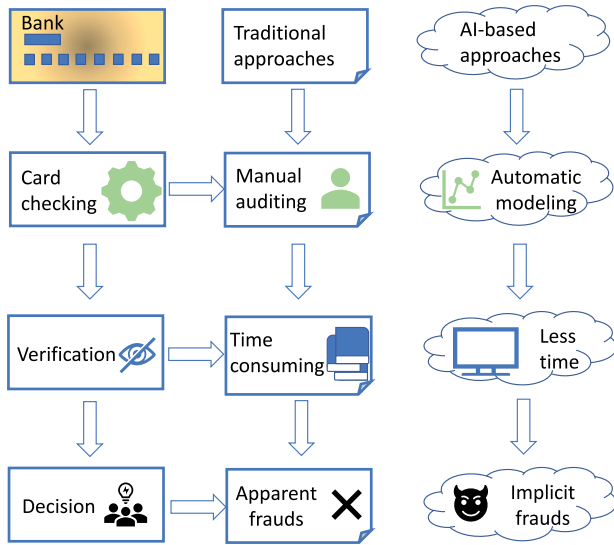


Fig. 1. Credit card fraud detection with traditional and AI methods. Traditional approaches are time-consuming and not precise, while AI-based methods are efficient and effective.

Conventional KL divergence generally overlooks the distribution ratios of data structures, thus causing the gradients to descend unstably. It may increase the hardness of model training.

Second, the current methods neglected the importance of features and did not mine useful information from them. Some data in realistic fraud detection may contain too much noise and too many useless features, and it could lead to the imperfection of prediction results.

Third, the layers of current attention-based methods are too shallow to capture enough characteristics of data. For example, LSTM-Attention [3] applies the combination of one-layer attention and two Long Short-Term Memory (LSTM) networks. The attention layer in the method does not exert its best capabilities to detect frauds. There is no significant difference between applying or not applying the layer. The formal experimental comparison is provided in Section 4.

To tackle the above issues, we propose a novel method named as **Balanced-Variational AutoEncoder-Attention (Bal-VAE-Attention)** which can significantly improve the existing fraud detection results.

In our method, we design a new oversampling method Bal-VAE to solve the first issue. We verify its performance through comparing with other existing oversampling methods given the same circumstances. It considers the imbalanced distribution of fraudulent and normal transactions in the designed new loss function. It surpasses the vanilla VAE and other oversampling methods under the condition of utilizing prevailing common machine learning models. We also provide solid mathematical proof. To address the second issue, we deploy Light Gradient Boosting Machine (LightGBM) model to conduct feature selection, which is capable of filtering useless features and obtains the most important embeddings for the latter inputting of main model structure. To resolve the third issue, we apply a deep multi-head attention structure that includes both encoders and decoders originating from Transformer [10]. Through substantial experiments and ablation study, we validate the effectiveness of our proposed method. We have it verified on two open-source credit card fraud datasets, and it surpasses other state-of-the-art baselines.

We summarize main contributions of our work below.

1. We propose a novel loss function and form a new oversampling method Bal-VAE with it. In this loss function, the imbalanced distribution of fraudulent and normal transactions is taken into consideration. We verify its performance theoretically and experimentally. Bal-VAE surpasses the vanilla VAE and other prevailing oversampling methods

given the same circumstances. We also provide solid mathematical proof in Appendix.

2. We adopt automatic feature selection to capture the most important features with representation learning. It successfully removes the features with low importance and reduces the noise in the original data.

3. We propose our main method Bal-VAE-Attention, which integrates Bal-VAE and attention mechanism to detect credit card frauds, and it deepens the layers of multi-head attention with masks. Our work achieves state-of-the-art results based on two open-source credit card fraud detection datasets.

The rest of our work is stated as follows. In Section 2, we concisely present and review related work. In Section 3, we elaborate and formalize our problem and methodology. We present our main experimental results and discussion in Section 4. We sum up our work in Section 5.

2. Related work

We introduce credit card fraud detection studies with augmentation techniques or with attention mechanism in the following subsections.

2.1. Credit card fraud detection with augmentation

Because of the severely imbalanced distribution of normal and fraud transactions, Generative Adversarial Networks (GANs) [11], Synthetic Minority Oversampling Technique (SMOTE) [12], and VAE [13] related algorithms are commonly applied in credit card fraud detection domain to act as oversampling methods. SMOTE is a traditional oversampling technique to handle data imbalance. Recently, GANs and VAE methods are booming, utilizing their generative abilities in simulating minority samples.

Researchers in [14] proposed an unsupervised feature learning method so as to boost the performance of multiple classifiers by deploying a stacked sparse autoencoder. This method showed superior performance over other baselines. In [15], a GANs-based method showed satisfactory performance in predicting frauds. This approach regularized the data generation process with loss functions to overcome the limitation of not capturing entire fraud space. Autoencoder and its variants showed extraordinary effects in fraud detection. In [9,16], an autoencoder and VAE structures were deployed as oversampling techniques. They were proven effective. Furthermore, the combination of sparse autoencoder and GANs were employed in [8], the authors first utilized sparse autoencoder to derive representations of normal transaction. Subsequently, they deployed a GAN and applied the discriminator of it to conduct detection. This method outperforms traditional state-of-the-art one class methods.

The above contributions are feasible. Nevertheless, the augmentation methods which most of them apply need improvement theoretically. Almost all of the current loss functions neglect the imbalanced distribution of normal and frauds, which could lead to the imperfection of detecting the specific fraudulent transactions. That is what we have tackled in our research.

2.2. Credit card fraud detection with attention

Attention mechanism was first introduced to optimize machine translation mission [10]. In recent years, plenty of researchers have employed it in credit card fraud detection.

The combination of attention modules with other neural network components enhances models' capabilities. In [3], an attention-based LSTM model was proposed to merge the advantages of three sub-models. It took the sequential characteristics of transactional data into consideration and detected the most important transaction that provides the highest accuracy related to the frauds. Scholars in [4] designed a similarity attention layer with BiLSTM, and it is able to measure similarity among data, assign weights to unique features,

Table 1
Key notations.

Symbol	Definition
X_i	Original inputting sequence
z	Latent variable
$p(X_i)$	Distribution function of X_i
m_i	Minor data sequence in X_i
n_i	Major data sequence in X_i
D_{KL}	Kullback–Leibler divergence
$p(z)$	Prior distribution of z
$p(x z)$	Distribution of x given z
$p(z x)$	Posterior distribution of z
$q(z x)$	Distribution approximating $p(z x)$
$E_{z \sim q(z x)}$	Expectation of z subject to $q(z x)$
x_{BAL}	Balanced variable of x
W_1, W_2	Weights for activation function ReLU
b_1, b_2	Biases for activation function ReLU
$LayerNorm()$	Normed neural networks layers
$Autoencoder()$	Neural networks for dimensionality reduction

and reduce overfitting in classification. Yu Xie et al. [6] proposed a time-aware-attention module and extracted the time-related behavioral information, which enables the model to capture temporal cardholders' behavior characteristics. Researchers in [17] depicted a model which integrates feature-wise attention and boosting techniques for fraud detection. It incorporated diversity regularization to the basic learners from the perspective of feature dimension.

Similarly, representation learning with attention layers and deep neural networks shows great detection abilities in other work. In [5], the 3D convolution layers were applied to integrate spatial and temporal representations of fraudulent patterns. In [7], sample-level and feature-level representations were integrated to form embeddings and to learn the entire sample space of details and feature dependencies.

All of the above contributions are proven effective, but they did not embody the full capabilities of attention mechanism. Their attention layers are too shallow to face up well with a more complex structure of transactional data, and this fact also inspires us to create our own work.

3. Methodology

We describe our method Bal-VAE-Attention in the following subsections. First, we formalize our research problem and explain the notations. Second, we elaborate on the feature selection, augmentation sketch, and attention structures, respectively. Finally, we depict our whole method pipeline and provide an algorithm explanation.

3.1. Problem formulation

We describe our research problem in (1).

$$P(F_i = 1 | IN_i, AF_i, p_i). \quad (1)$$

In the above equation, F_i stands for the i th binary classification results for fraud detection, where 1 denotes fraudulent transaction, and 0 means normal one. IN_i stands for the basic transaction information of a normal user. AF_i represents the augmented feature information for frauds, and p_i is a positional embedding that controls the maximum length of an inputted sequence. The function $P()$ denotes the conditional probability of the whole expression.

3.2. Notations statement

In this subsection, we provide detailed definitions in Table 1 for important symbols that will appear in the latter principles and algorithm.

3.3. Feature selection

We apply LightGBM [18] as the tool to conduct an automatic feature selection process.

LightGBM, developed by Microsoft in 2017, is a tree-based gradient-boosting framework. Its most extraordinary advantages are efficiency and distributed computing. There are built-in functions that can calculate feature importance scores automatically and visualize the sortation of importance.

We first train the vanilla LightGBM model on our datasets. Then, we calculate the feature importance scores based on information gain feature importance method. The gain quantifies each feature's contribution to the model by assessing the total gain when that feature is utilized in decision trees. A higher gain value indicates greater importance for making predictions. It measures how much a feature enhances the model's predictions, specifically by evaluating the quality of the splits it produces. After that, the sortation can be performed automatically. As a consequence, we derive the needed features with the top five highest feature importance scores. We plot the importance ranking results according to the feature selection process.

3.4. Framework for data augmentation

In this subsection, we revisit VAE [13] and provide key components of the newly proposed version Bal-VAE.

3.4.1. The vanilla VAE

VAE is a neural network architecture to generate synthetic samples. It applies sufficient and explicit regularization in the latent space compared with conventional Autoencoders. The basic framework of VAE is illustrated in Fig. 2.

The core components of VAE are an encoder and a decoder. The encoder obtains input data and transforms them into dense representations while the decoder converts the representations back into original forms with minimization of loss. Through this procedure, it generates outputs. During the loss optimization step, a key notation, reconstruction loss, needs elaboration. It denotes the mean-squared error between input X_i and output $\hat{X}_i$. The general training steps of a typical VAE include the following procedures.

- It encodes inputs as distribution rather than points.
- We sample a point of the latent space from that distribution.
- The latent layer performs regularization.
- We decode the sampled point and calculate the reconstruction loss.
- The reconstruction loss is backpropagated through the network and the overall method loss is minimized.

The whole structure embodies the dimension reduction principles and generative capabilities of a neural network. The vital procedure of this method is to figure out a proper latent layer to approximate the original distribution.

In order to obtain a latent variable Z that is similar to X_i , we need to introduce a posterior distribution function $p(z|x)$. In this structure, $q(z|x)$ is introduced to approximate $p(z|x)$. Therefore, we need to minimize the KL Divergence of $q(z|x)$ and $p(z|x)$.

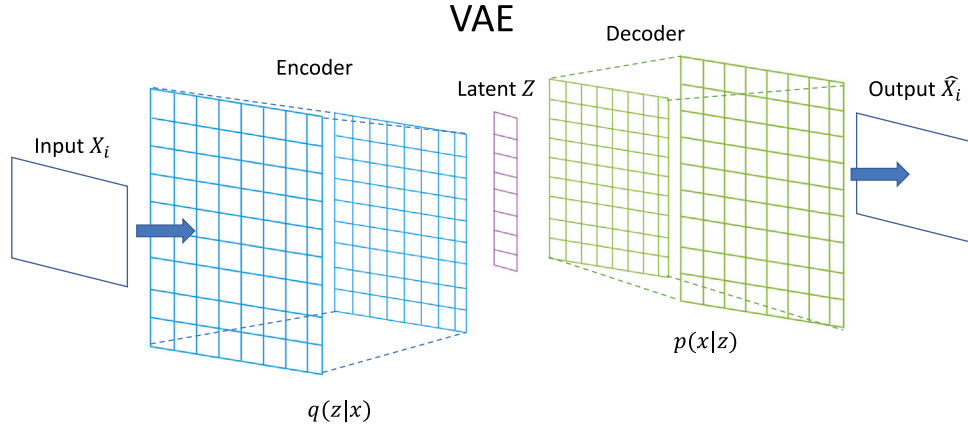
This KL Divergence can be computed through Eq. (2).

$$D_{KL}[q(z|x)||p(z|x)] = E_{z \sim q(z|x)}[\log q(z|x) - \log p(z|x)]. \quad (2)$$

By applying Bayesian theorem, we derive Eq. (3).

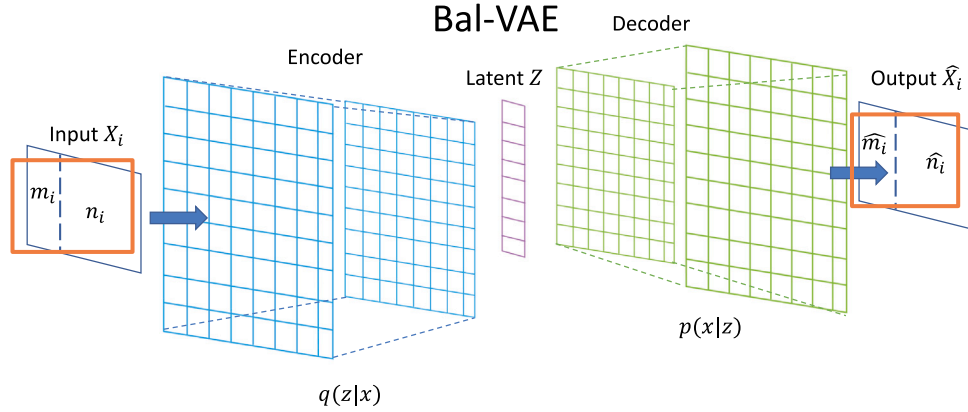
$$\begin{aligned} \log p(x) - D_{KL}[q(z|x)||p(z|x)] &= E_{z \sim q(z|x)}[\log q(x|z)] \\ &\quad - D_{KL}[q(z|x)||p(z)]. \end{aligned} \quad (3)$$

In Eq. (3), $\log p(x)$ denotes the overall objective we need to optimize. Since $q(z|x)$ is utilized to approximate $p(z|x)$, $D_{KL}[q(z|x)||p(z|x)]$ is close to zero. The first and second terms represent reconstruction loss



$$\text{Loss} = l_{\text{reconstruction}}(x, \hat{x}) + \sum D_{KL}[q(z|x)||p(z)]$$

Fig. 2. Framework of VAE. The encoder transforms the input data into dense representations while the decoder converts the representations back into original forms with minimization of loss.



$$\text{NewLoss} = l_{\text{reconstruction}}(x, \hat{x}) + \sum D_{KL}[q(z|x_{BAL})||p(z)]$$

Fig. 3. Framework of Bal-VAE. The rectangles symbolize newly built structures. They consider the imbalanced ratios of input data.

and regularization term, respectively. Because of the gradient descent principle, we need to minimize $-p(x)$. This problem is equivalent to maximize $E_{z \sim q(z|x)} \log p(x|z) - D_{KL}(q(z|x)||p(z))$, thus leading to the minimization of $D_{KL}(q(z|x)||p(z))$.

3.4.2. Bal-VAE

We propose Bal-VAE as the data augmentation method based on the vanilla VAE. We create a novel loss function and it considers the imbalanced ratio of data. The overview of the augmentation procedure is visualized in Fig. 3.

Similar to the vanilla VAE, the essence of this structure is to generate samples that are similar to original ones. The main difference is we incorporate imbalanced ratios in the design. Through applying this method, the novel loss function will pay balanced attention to the original imbalanced data, and it will enhance the ability to detect the minority. The inputting X_i sequences denote the original inputting features of the objective needing to be enhanced, i.e., the minority. Here, we define the parameters of minor data objects as m_i and the parameters of majority as n_i . X_i sequences are transformed in an Encoder structure. Then, the latent variable Z is set to represent the controlling factors of the inputs. After mapping Z into a Decoder structure, we derive the distribution function $p(X_i)$ and obtain the output $\hat{X}_i$ with the estimated parameters $\hat{m}_i$ and $\hat{n}_i$.

We propose detailed definitions and Theorem 3.1 as follows.

Definition 3.1 (L1). Reconstruction Loss-L1 denotes the decoder to sample latent variable Z that most approximates X_i .

$$L1 = E_{z \sim q(z|x)} \log p(x|z). \quad (4)$$

Because we can assume the prior distribution $p(z)$ obeys Gaussian distribution $N(0, 1)$ for simplicity, and $q(z|x)$ obeys $N(\mu_{(i)}(x), \sigma_{(i)}^2(x))$, the vanilla KL loss function is essentially the KL divergence between $N(\mu_{(i)}(x), \sigma_{(i)}^2(x))$ and $N(0, 1)$. Thus, we have Definition 3.2.

Definition 3.2 (L2). Vanilla KL Loss Function-L2 denotes the regularization term for the target problem.

$$\begin{aligned} L2 &= D_{KL}(q(z|x)||p(z)) \\ &= \frac{1}{2} \sum (\mu_{(i)}^2(x) + \sigma_{(i)}^2(x) - \ln \sigma_{(i)}^2(x) - 1). \end{aligned} \quad (5)$$

where $\mu(x)$ denotes the mean of variable x , and $\sigma^2(x)$ represents the variance of x .

The symbol x_{BAL} represents the balanced variable of original input x . It considers the distribution ratios of majority and minority. Through

applying these ratios as the coefficients to each term of $L2$, we obtain a new loss function $L3$, which can be expressed in Definition 3.3.

Definition 3.3 ($L3$). New KL Loss Function- $L3$ denotes the imbalanced regularization term for the target problem.

$$\begin{aligned} L3 &= D_{KL}(q(z|x_{BAL})||p(z)) \\ &= \frac{1}{2} \sum (\ln(\frac{n(m_i)}{n(n_i)}) \mu_{(i)}^2(x) \\ &\quad + (\frac{n(m_i)}{n(n_i)}) \ln \sigma_{(i)}^2(x) \\ &\quad - \frac{1}{2} \ln(\frac{n(n_i)}{n(m_i)}) \ln \sigma_{(i)}^2(x) - 1). \end{aligned} \quad (6)$$

where $n(m_i)$ indicates the number of minority, $n(n_i)$ indicates the number of majority, $n(n_i) \gg n(m_i)$, $\mu(x)$ stands for the mean of variable x , and $\sigma^2(x)$ represents the variance of x . $L3$ mainly incorporates the ratios of minority and majority as the coefficients of each term in vanilla loss function $L2$. Through applying this adjustment, the new loss function $L3$ will pay more attention to the imbalanced distribution of original data.

Definition 3.4 ($L4$). Overall NewLoss- $L4$ denotes the whole loss function that considers data imbalance.

$$L4 = L1 + L3. \quad (7)$$

Theorem 3.1 (*Loss Function Theorem*). Given original KL Divergence Loss function $L2$ and new KL Divergence Loss function $L3$, $L3 = o(L2)$.

Proof. See detailed derivation in Appendix. $\square$

3.5. Multi-head attention-based structure

Multi-Head Attention was proposed in [10] to optimize machine translation missions initially. It gradually revolutionized nearly all of the domains in computer science and AI. The basic principle of multi-head attention is that it allows parallel processing of projected queries, keys, and values. It has a structure called a head, which denotes the number of attention pooling outputs.

In our method, we attach three deep masked sequential multi-head attention to the encoder and decoder layers of the whole model. We use mask mechanism to hide illegal and disruptive connections among sequences. In particular, the mask operates on an inputting matrix and returns only the upper triangular part of it, thus preventing access to future positions and permitting the model only to consider past and present positions when making predictions. The masked positions allow the model to conduct attention mechanism for all tokens in a batch simultaneously without the need for sequential processing. Therefore, the whole process makes the training procedures parallel.

3.6. Overall method pipeline

We describe our whole method structure in Fig. 4. It mainly contains three stages. First, we extract useful features from the original inputting. Second, we enhance the training data to achieve a balanced ratio of normal transactions and frauds with Bal-VAE. Third, we apply multi-head attention to further deepen the neural network layers.

In detail, the principles of attention mechanism essentially cover different weights or importance of input sequences' certain parts. It generates attention scores with scaled dot products consisting of queries, keys, and values. Multi-head attention mainly focuses on the design of a head, which allows the model to capture diverse information and compute simultaneously. As to the multi-head attention structures in our design, we first embed the input sequence into encoder and decoder dimensional space. The encoder consists of position and feature embeddings and the decoder contains position and response embeddings.

Algorithm 1 Bal-VAE-Attention Pipeline.

Input: E : Dataset; D : latent dimension; N : the inputting of the decoder; H : number of features in batchnorm layer.

```

1: FEATURE EXTRACTION:
2:    $E_{new} \leftarrow E_{important\ features} + labels$ 
3: AUGMENTATION:
4:   Split  $E_{new}$  into training set  $E_{train}$  and testing set  $E_{test}$ .
5:    $Fraud \leftarrow E_{train}[label' == 1]$ 
6:   for  $X_i$  sampled from  $Fraud$  do
7:      $X_j \leftarrow Autoencoder(D, H, X_i)$ 
8:     Calculate Loss Function  $L$  via Equ. (7).
9:      $\hat{X}_i \leftarrow Bal - VAE(X_j)$ 
10:     $E_{trainnew} \leftarrow E_{train}, \hat{X}_i$ 
11:   end for
12: ATTENTION:
13: ENCODER:
14:    $NET \leftarrow ReLU(jW_1 + b_1)W_2 + b_2$ 
15:    $M \leftarrow E_{trainnew}$ 
16:    $E_1 \leftarrow M + Multihead(LayerNorm(M, M, M))$ 
17:    $E_2 \leftarrow E_1 + NET(LayerNorm(E_1))$ 
18: DECODER:
19:    $D_1 \leftarrow N + Multihead(LayerNorm(N, N, N))$ 
20:    $D_2 \leftarrow D_1 + Multihead(LayerNorm(D_1, E_2^{enc}, E_2^{enc}))$ 
21:    $D_3 \leftarrow D_2 + NET(LayerNorm(D_2))$ 
Output:  $\hat{Y} \leftarrow FC(D_3(E_{test}))$ 

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Then, the input is linearly projected into different spaces for each head, creating separate query, key, and value matrices. We create multiple attention layers and update these matrices. After stacking the encoder, decoder, and multi-head attention structures, we add a fully connected layer to the output of decoder layers, deriving the final prediction.

In particular, the testing set remains unchanged for the sake of ensuring the truth of model performance. The pipeline is provided as Algorithm 1.

During our whole pipeline, we first conduct automatic feature extraction, select the most important features, form integrated embeddings, and preprocess the data. In the augmentation stage, we split the dataset and enhance the fraudulent training data with Bal-VAE method by applying the Autoencoder() and novel loss function we built. After augmentation, we stack deepened masked multi-head attention mechanism.

In the multi-head attention structure of the algorithm, we design two core components: the encoder layers and decoder layers. Note that the encoder layers consist of positional embedding and extracted feature embeddings, and the decoder layers employ positional and response embedding related to the labels. During the encoder construction process, we first utilize NET layer to form a feed forward network that maps and normalizes the attention components. It consists of a $ReLU$ activation function with weights and biases. Then, we construct a transitional encoder component E_1 with normalized training input sequences and multi-head attention structure. We obtain E_2 with activated and normalized E_1 as the output of encoder. At the decoder building stage, we use multi-head attention operation as well as similar layer normalization to derive the first transitional encoder variable D_1 . Moreover, the former encoder output E_2 is incorporated into D_1 to form the second transitional decoder variable D_2 . After that, we utilize similar NET and layer normalization operation to gain D_3 as the final output of decoder structure. We derive the prediction results $\hat{Y}$ after linking a fully connected layer to D_3 with testing set.

4. Experiments

In this section, we elaborate on experimental procedures and comparison. We depict the applied datasets, evaluate specific method results, and discuss our work.

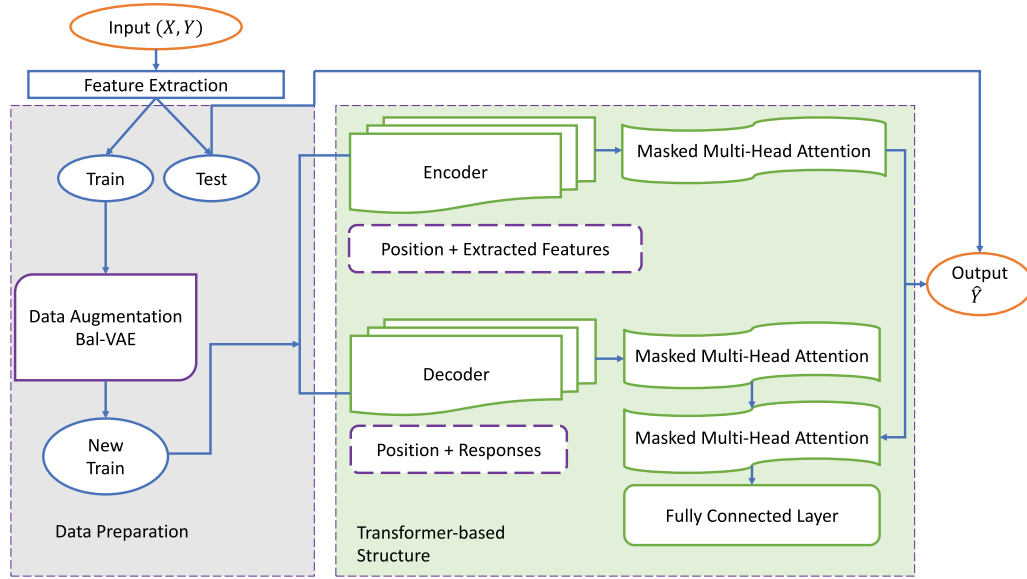


Fig. 4. Overview of Bal-VAE-Attention. It consists of data preparation and an attention-based framework. Firstly, we conduct feature extraction and dataset splitting. Secondly, we apply the proposed Bal-VAE method as augmentation of training set. Then, we import the enhanced training set into masked multi-head deep attention structures and derive the final prediction with testing set.

Table 2
European credit card fraud dataset.

Indicator	Content
Number of instances	284 807
Number of frauds	492
File size	150.83 MB
Features summary	Time, V1, V2, V3, ..., V28, Amount
Label	Class

Table 3
Simulated credit card fraud dataset.

Indicator	Content
Number of instances	87 506
Number of frauds	7506
File size	351.24 MB
Features summary	Trans_time, cc_num, ..., merch_long
Label	is_fraud

4.1. Experimental setup

We implemented all the experiments on a Linux server and a MacBook. The server contains an NVIDIA RTX A6000 Graphic Card with 48G as its memory. The devices of the MacBook are a 1.2 GHz Intel Core m3 CPU, 8 GB LPDDR3 and an Intel HD Graphics 615 GPU. We programmed with Python 3.9, Scikit-learn machine learning library, TensorFlow framework, and PyTorch Lightning framework with CUDA 11.7.

4.2. Datasets

We apply two open-source datasets in our research. No missing data are detected in both datasets.

The first dataset is collected by the Machine Learning Group of ULB (Université Libre de Bruxelles) and Worldline. At present, it is an open-source dataset available on Kaggle.¹ The dataset contains transactions executed by European credit cardholders in September 2013. In this dataset, there are 492 frauds out of 284,807 transactions, and it is highly imbalanced. The positive ratio is 0.173%. We present a detailed description in Table 2.

Note that the features “V1” to “V28” are anonymous indicators pre-processed with Principal Component Analysis (PCA) for the sake of data privacy. “Time” and “Amount” record the transactional information for each instance. All of the 30 features belong to numerical data. “Class” denotes the label for each transaction where “0” means a legitimate one and “1” denotes a fraudulent one.

The second dataset is a simulated credit card fraud detection dataset available on Kaggle.² It contains legal and fraudulent transactions from the duration of 1st Jan 2019 to 31st Dec 2020. It encompasses transactional and demographic information of 1000 credit cardholders’ transactions with a pool of 800 merchants. We took 80 000 normal transactions randomly with 7506 fraudulent ones. The positive ratio is 8.58%. The detailed description of the dataset is stated in Table 3.

The original features contain 22 text and numeric variables including transactional time and index, merchant information, names and gender of cardholders, etc. We deploy transformation and filtering to cleanse the data. In detail, we use “fit_transform” function in Scikit-learn Python library to transform textual and categorical information into numerical forms. Moreover, we delete useless demographic data such as the addresses of cardholders. Finally, we convert the versatile feature variables into 14 float variables. The label “is_fraud” is composed of “0” and “1”, where “0” means a legitimate transaction and “1” represents a fraudulent one.

4.3. Baselines

4.3.1. Augmentation baselines

We describe the principles of deployed augmentation baselines below.

- SMOTE [12] is a general oversampling method for imbalanced data. This algorithm starts with figuring out k nearest-neighbors of minority based on Euclidean Distance, then picks one randomly and produces a sample similar to it.

¹ <https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud>.

² <https://www.kaggle.com/datasets/kartik2112/fraud-detection>.

- GAN [11] has two main ingredients: a discriminator and a generator. Through training, the generator tries to fool the discriminator and produces samples that cannot be detected as fake ones. Finally, both structures achieve equilibrium and the probability of detecting a fake sample by the discriminator is 0.5.
- VAE [13] has an encoder–decoder structure. The encoder imports the original variables, and the latent variable extracts representation and maps it to the decoder structure. Finally, the decoder reconstructs similar variables.

4.3.2. Traditional machine learning baselines

Conventional machine learning approaches for fraud detection are listed as follows.

- Random Forest (RF) [19] is an ensemble decision tree-based approach. It consists of multiple trees and exports the mode of their classification or regression results.
- LightGBM [18] belongs to the gradient boosting methods and applies Exclusive Feature Bundling (EFB). It is mainly based on weak decision tree models and tries to figure out the best split of leaves through iterations. One of its advantages is its fast computing speed and light structure.
- eXtreme Gradient Boosting (XGBoost) [20] is a sparsity-aware algorithm for sparse data and a weighted quantile architecture for approximate tree learning. It designs an exact greedy algorithm to find the best split. Its running speed is extremely high and costs few computational resources.

4.3.3. Credit card fraud detection baselines

We briefly depict several state-of-the-art credit card fraud detection baselines in the following items.

- LSTM-Attention [3] integrates three sub-models: the LSTM networks which incorporate transactional sequences, the uniform manifold approximation and projection (UMAP) to derive the most valuable predictive features, and the attention mechanism which is capable of improving LSTM performance.
- DevNet [21] applies an end-to-end learning of anomaly scores by a neural deviation learning, where some labeled anomalies and a prior probability to execute statistically significant deviations are leveraged.
- SAE+GAN [8] delivers a method that combines sparse autoencoder (SAE) and GAN. It first derives representations of normal transactions and then trains a GAN with them. By incorporating SAE and GAN, the model can determine the genuineness of a transaction.
- TAI-LSTM [6] designs a time-aware and attention-based method to extract transactional behavioral representations of cardholders. The attention module enables the model to integrate a user's past and present transactional details. Together with modified LSTM structures, the model can capture both long-term and short-term transactional variations.
- GAN-TabularAttention [22] employs a GAN structure to generate fraudulent classes to overcome data imbalance. A multi-head attention mechanism is further connected to strengthen the learning capability of the model.
- UAD-FDNet [23] incorporates a channel-wise feature attention module with a GAN structure. The GAN is trained to mimic normal transactions. Thus, it will be sensitive to discriminate fraudulent testing samples from normal ones. Moreover, the attention module is added to the generator of GAN to improve its learning abilities regarding reconstruction of trained samples.

4.4. Evaluation metrics

Since the common metric “accuracy” cannot fully express the model performance under the imbalanced distribution condition, we deploy metrics that focus on the ability to detect positive samples.

We introduce Recall, Precision, AUC-ROC, and AUC-PR as our metrics.

Recall stands for the ratio of correctly predicted positive samples to the whole positive samples. It can be denoted as Eq. (8).

$$Recall = \frac{RP}{RP + FN} \quad (8)$$

where RP denotes real positive, FN denotes false negative.

Precision denotes the quantity of real positive samples in all the predicted true samples. The metric can be formalized in Eq. (9).

$$Precision = \frac{RP}{RP + FP} \quad (9)$$

where FP means the false positive.

AUC-ROC means the area under the Receiver Operating Characteristic (ROC) Curve, which shows a model's capabilities to distinguish between positive and negative samples. Given predicted probabilities of negative or positive results of each sample, we rank the samples from the lowest probabilities to the highest ones. AUC-ROC is equivalent to the probability that a randomly chosen positive case is ranked higher than a randomly chosen negative one [24]. The larger the AUC-ROC value, the more likely it is that positive samples will be ranked in front of negative samples. We provide Eq. (10) as one of its computing formulas.

$$AUC-ROC = \frac{\sum_{n \in positive} rank_n - \frac{P(1+P)}{2}}{PQ} \quad (10)$$

where P represents the quantity of positive samples, Q denotes the number of negative ones, $rank_n$ represents the ranking result of the n th positive sample based on predicted probabilities.

AUC-PR denotes the area under the Precision–Recall (PR) Curve. It is the average of precision and recall given a threshold of certain probability. It is in contrast to AUC-ROC in that the axes of a PR curve are the recall and precision, while the y -axis of ROC Curve is the recall and the x -axis is False Positive Rate (FPR).

4.5. Configurations

We implement two sets of experiments. In the first set, we deploy our novel oversampling method and verify its effectiveness with trivial machine learning approaches. In the second set, we apply our whole method and compare its performance to other state-of-the-art baselines.

In the first round of experiments, we utilize TensorFlow and PyTorch framework. We split the dataset into training, validation and testing sets according to the ratio of 6:2:2. We select Adam as our optimizer and employ our new loss function combined with Binary CrossEntropy.

At the second stage, we deploy StratifiedShuffleSplit method to split the data with 1:3 as the ratio of testing and training sets in each fold, which can ensure the balanced distribution of both label types in each batch. We set the early stop value as 5 rounds so as to prevent overfitting. Also, we set the batch size as 128, the maximum sequential length for an episode of inputting as 500, the embedding dimension as 512 and the number of heads for encoder and decoder as 8. Through fine-tuning of the whole model, we figure out this hyperparameter combination can be the most proper setting and is capable of achieving the best performance.

4.6. Results

We deliver our main experimental results in the following subsections.

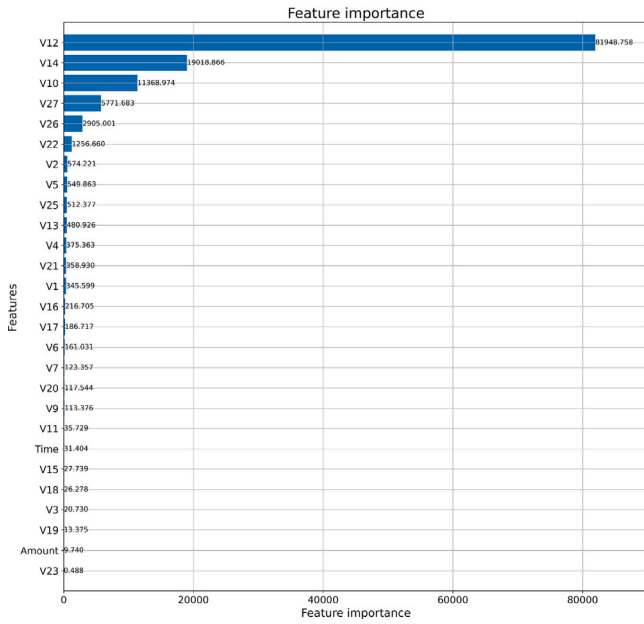


Fig. 5. Feature importance of European dataset. It presents the rank of features from the most significant ones to the least important ones.

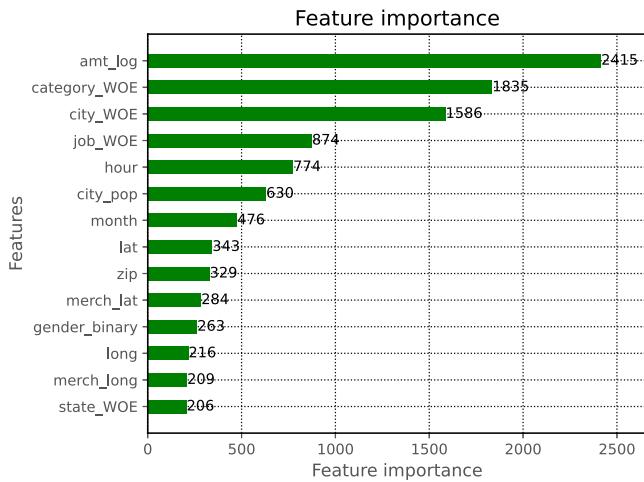


Fig. 6. Feature importance of synthetic dataset. It displays similar ranking results.

4.6.1. Feature extraction

To begin with, we conduct normalization of features. Then, we pick out automatically the most important features by applying LightGBM, and plot the results regarding the first dataset in Fig. 5.

As shown in Fig. 5, the features “V12”, “V14”, “V10”, “V27”, “V26” have the most significance, and other ones are negligible. Therefore, the above features are imported into the next stage.

As to the second dataset, We plot the results in Fig. 6.

We select “amt_log”, “category_WOE”, “city_WOE”, “job_WOE”, and “hour” as the inputs to our model because of their high importance.

4.6.2. Augmentation methods

We conduct comparison in the first round of experiments. We apply our augmentation method and compare it with SMOTE, GAN, and the vanilla VAE. Dataset 1 denotes the European credit dataset, and Dataset 2 denotes the synthetic one.

Firstly, we train the augmentation method and plot the training procedure as Fig. 7.

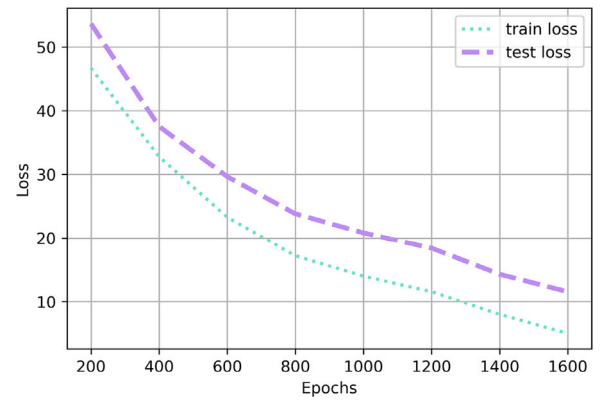


Fig. 7. Loss of Bal-VAE distribution. Both training and testing losses decrease and converge in the end.

Table 4

Augmentation methods with Random Forest.

Datasets	Metrics	Method				
		RF	RF+SMOTE	RF+GAN	RF+VAE	RF+Bal-VAE
Dataset 1	Precision	0.8873	0.2403	0.8837	0.5662	0.8415
	Recall	0.6774	0.8710	0.4086	0.8280	0.7742
	AUC-ROC	0.8386	0.9332	0.7043	0.9135	0.8870
	AUC-PR	0.6016	0.2096	0.3621	0.4691	0.6485
Dataset 2	Precision	1	0.1627	0.9696	0.8521	0.9625
	Recall	0.0565	0.8839	0.6072	0.7386	0.7450
	AUC-ROC	0.5282	0.7169	0.8027	0.8629	0.8730
	AUC-PR	0.1414	0.1543	0.6241	0.6529	0.6101

Fig. 7 shows that the training and testing loss decreases rapidly and converges after 1600 epochs, and it proves that the method is effective.

Then, we select RF, LightGBM, and XGBoost as our baselines to show its effect given the same conditions. The results are shown in Table 4, Table 5, and Table 6.

In Table 4, RF+Bal-VAE has the highest AUC-PR regarding the first dataset, and all the metrics are evenly high enough to show its prediction effect. Although the other three augmentation methods have one metric extremely high, they all have defection, i.e., one metric among the four is too low, and this fact enhances the capability of our proposed augmentation method. In particular, the recall value of random forest only model is extremely low because of its low capability to generate positive examples. In both datasets, the augmentation structure RF+Bal-VAE shows stable and high performance.

In Table 5, LightGBM+Bal-VAE has much better results than the other methods regarding Dataset 1. Its Precision, AUC-ROC, and AUC-PR are all the highest. Although the vanilla LightGBM has the highest recall value, its precision is too low. As to Dataset 2, LightGBM+Bal-VAE also shows its superiority in general.

In Table 6, the method XGBoost+Bal-VAE has the highest Recall value regarding Dataset 1, and the other three metrics are also evenly high, which is similar to RF+Bal-VAE. Also, the other three augmentation methods have one or two metrics that are not high enough. In Dataset 2, all of the models show high performance, but XGBoost+Bal-VAE method is also proven to be superior to other baselines.

All these facts show that our proposed augmentation method has strong and stable performance, regardless of the difference in prediction baselines and datasets. In particular, it significantly surpasses the vanilla VAE method regarding model capability.

4.6.3. Overall methods comparison

We conduct the second round of experiments and verify the performance of the whole method. We reproduce all the baselines and present the comparison results in Table 7.

Table 5
Augmentation methods with LightGBM.

Datasets	Metrics	Method				
		LightGBM	LightGBM+SMOTE	LightGBM+GAN	LightGBM+VAE	LightGBM+Bal-VAE
Dataset 1	Precision	0.2068	0.4937	0.1229	0.4321	0.9024
	Recall	0.8495	0.8387	0.7742	0.7527	0.8065
	AUC-ROC	0.9336	0.9244	0.9489	0.9344	0.9733
	AUC-PR	0.4255	0.6753	0.0651	0.2451	0.8186
Dataset 2	Precision	0.9560	0.8797	0.9630	0.8554	0.9356
	Recall	0.8634	0.8815	0.8153	0.7010	0.6676
	AUC-ROC	0.9953	0.9879	0.9936	0.8452	0.8317
	AUC-PR	0.9694	0.9431	0.9623	0.6242	0.6518

Table 6
Augmentation methods with XGBoost.

Datasets	Metrics	Method				
		XGBoost	XGBoost+SMOTE	XGBoost+GAN	XGBoost+VAE	XGBoost+Bal-VAE
Dataset 1	Precision	0.8889	0.0567	0.9103	0.8065	0.8065
	Recall	0.7742	0.8925	0.7634	0.7075	0.9342
	AUC-ROC	0.9529	0.9626	0.9885	0.9302	0.9700
	AUC-PR	0.8262	0.3269	0.8457	0.6799	0.8308
Dataset 2	Precision	0.9375	0.9342	0.9415	0.9409	0.9467
	Recall	0.9199	0.9303	0.9310	0.9206	0.9282
	AUC-ROC	0.9572	0.9622	0.9629	0.9577	0.9618
	AUC-PR	0.8689	0.8748	0.8822	0.8727	0.8846

Table 7
Overall methods comparison.

Datasets	Method	Recall	Precision	AUC-ROC	AUC-PR
Dataset 1	LSTM-Attention	0.9191 $\pm$ 0.01	0.9885 $\pm$ 0.01	0.9653 $\pm$ 0.01	0.7403 $\pm$ 0.01
	LSTM-(Attention)	0.8858 $\pm$ 0.01	0.9885 $\pm$ 0.01	0.9842 $\pm$ 0.01	0.7867 $\pm$ 0.02
	DevNet	0.8424 $\pm$ 0.01	0.8424 $\pm$ 0.01	0.9600 $\pm$ 0.02	0.6500 $\pm$ 0.02
	SAE+GAN	0.7937 $\pm$ 0.01	0.9759 $\pm$ 0.01	0.99 $\pm$ 0.01	0.99 $\pm$ 0.01
	TAI-LSTM	0.99 $\pm$ 0.01	0.5070 $\pm$ 0.01	0.9380 $\pm$ 0.001	0.7060 $\pm$ 0.02
	GAN-TabularAttention	0.3490 $\pm$ 0.01	0.99 $\pm$ 0.01	0.99 $\pm$ 0.01	0.4523 $\pm$ 0.02
	UAAD-FDNet	0.7553 $\pm$ 0.01	0.9795 $\pm$ 0.01	0.9940 $\pm$ 0.002	0.7983 $\pm$ 0.02
	Bal-VAE-Attention (Ours)	0.9990 $\pm$ 0.001	0.9991 $\pm$ 0.0009	0.9995 $\pm$ 0.0005	0.9998 $\pm$ 0.0002
Dataset 2	LSTM-Attention	0.8177 $\pm$ 0.01	0.9153 $\pm$ 0.01	0.9184 $\pm$ 0.02	0.7549 $\pm$ 0.01
	LSTM-(Attention)	0.8105 $\pm$ 0.01	0.9152 $\pm$ 0.01	0.9004 $\pm$ 0.02	0.7564 $\pm$ 0.02
	DevNet	0.0261 $\pm$ 0.005	0.0261 $\pm$ 0.005	0.5130 $\pm$ 0.005	0.0905 $\pm$ 0.01
	SAE+GAN	0.1716 $\pm$ 0.01	1 $\pm$ 0.00	0.1716 $\pm$ 0.01	0.5848 $\pm$ 0.001
	TAI-LSTM	0.9320 $\pm$ 0.01	0.9928 $\pm$ 0.005	0.9702 $\pm$ 0.001	0.9662 $\pm$ 0.02
	GAN-TabularAttention	0.4986 $\pm$ 0.01	0.99 $\pm$ 0.01	0.7241 $\pm$ 0.01	0.5742 $\pm$ 0.02
	UAAD-FDNet	0.99 $\pm$ 0.01	0.3189 $\pm$ 0.01	0.5 $\pm$ 0.01	0.6595 $\pm$ 0.02
	Bal-VAE-Attention (Ours)	0.99 $\pm$ 0.01	0.99 $\pm$ 0.01	0.99 $\pm$ 0.01	0.99 $\pm$ 0.01

^aLSTM-(Attention) indicates LSTM only structure.

Table 7 proves that our model exceeds other fraud detection baselines in performance regarding both datasets. The metrics that are close to 100% show the magnificent power of the combination of data augmentation and attention mechanism. Generally, some methods can be as effective as ours regarding one or two indicators. However, our approach is more stable and balanced. LSTM-(Attention) shows slight difference in performance compared to the vanilla LSTM-Attention. For example, both have the same Precision value of 0.9885 ± 0.01 regarding Dataset 1. This fact suggests the insufficient deploying of the attention layer in this method.

Moreover, we plot the related ROC Curves and PR Curves covering two datasets in Figs. 8, 9, 10 and 11.

Fig. 8 demonstrates the high performance of our method. It is the closest to the y-axis, which achieves a perfect AUC-ROC score.

The principle of PR Curve is that a model has better effect if it is closer to the upper right corner. Fig. 9 indicates that our model has the highest performance.

Similarly, we obtain ROC Curves regarding Dataset 2 in Fig. 10. The visualization demonstrates the superior performance of our method.

We derive similar results as shown in Fig. 11.

Note that the plotting and numeric results indicate some methods have suboptimal effects on Dataset 2. The possible reasons cover the

network frames of these approaches might not meet the structure of the simulated dataset. Due to the difference in characteristics between synthetic and real data, the average performance of these approaches can vary. Nevertheless, our Bal-VAE-Attention approach overcomes this disparity and performs steadily. All these facts prove the effectiveness of our proposed method.

4.6.4. Ablation study

We conduct sufficient ablation studies to interpret our method. The results are delivered in Tables 8 and 9.

Table 8 suggests the effect of removing key structures of our method regarding Dataset 1. We can derive that augmentation is necessary because no true positive samples are detected in the method that does not apply augmentation. Moreover, after we delete the encoder or decoder in the method, the model shows apparent defection. All these facts prove the effectiveness of the main ingredients of our whole method.

Table 9 suggests the similar results regarding Dataset 2. In summary, all these ablation study results prove the necessity of introducing augmentation and deep attention structures.

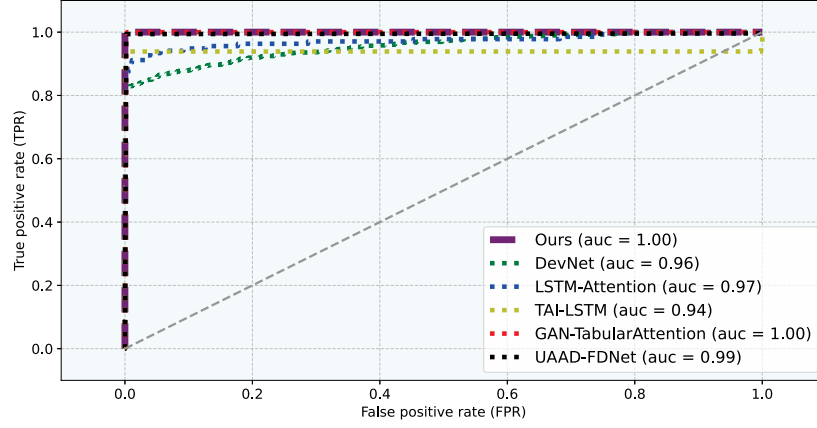


Fig. 8. ROC Curves of our models and baselines regarding Dataset 1. All of the models have high AUC-ROC values on Dataset 1. Our method outperforms the others.

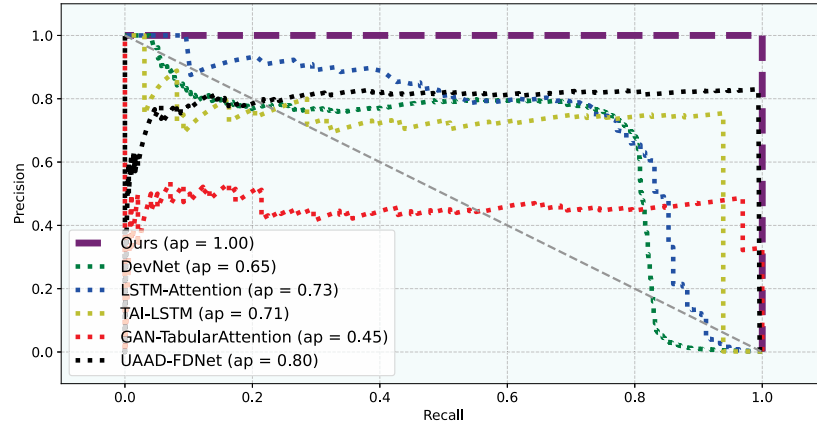


Fig. 9. PR Curves of our models and baselines regarding Dataset 1. Our method has the highest AUC-PR value.

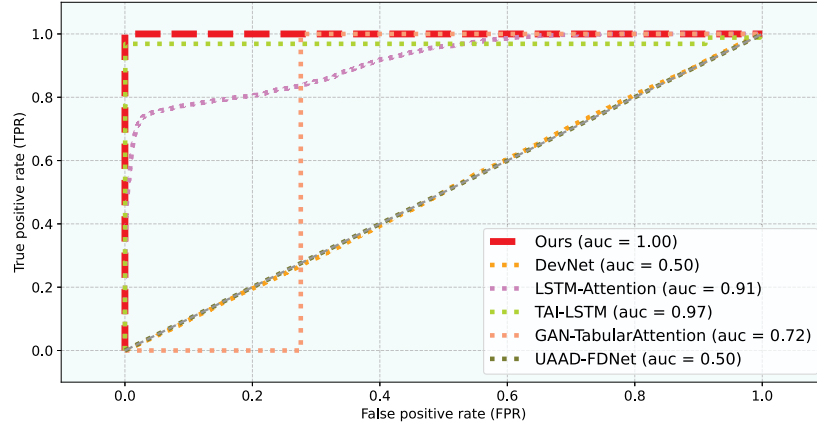


Fig. 10. ROC Curves of our models and baselines regarding Dataset 2. Our method ranks first.

Table 8
Ablation study Results-Dataset 1.

Method		Recall	Precision	AUC-ROC	AUC-PR
No augmentation		0 ± 0.00	0 ± 0.00	1 ± 0.00	1 ± 0.00
With augmentation	Encoder only	0.1663 ± 0.01	0.5009 ± 0.01	0.5007 ± 0.01	0.5005 ± 0.01
	Decoder only	0.1157 ± 0.01	0.9726 ± 0.01	0.5773 ± 0.01	0.6224 ± 0.01
Bal-VAE-Attention (Ours)		0.9990 ± 0.001	0.9991 ± 0.0009	0.9995 ± 0.0005	0.9998 ± 0.0002

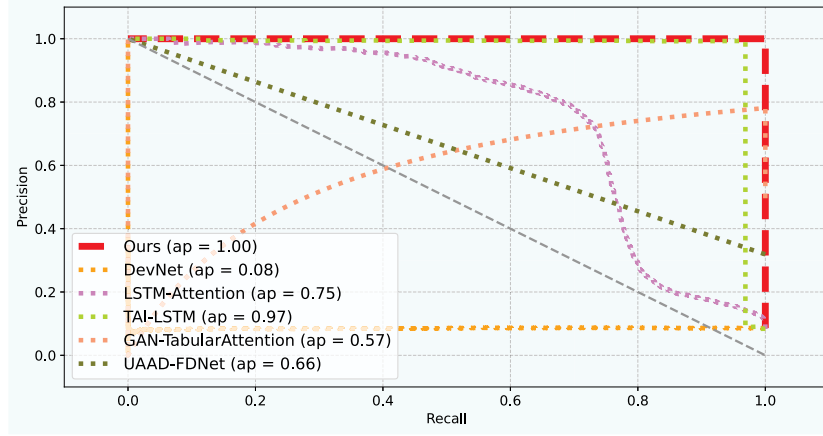


Fig. 11. PR Curves of our models and baselines regarding Dataset 2. Our method ranks first.

Table 9
Ablation study Results-Dataset 2.

Method	Recall	Precision	AUC-ROC	AUC-PR
No augmentation	0 ± 0.00	0 ± 0.00	0.002136 ± 0.001	0.06213 ± 0.001
With augmentation	Encoder only	0.5396 ± 0.01	0.1016 ± 0.01	0.5873 ± 0.01
	Decoder only	0 ± 0.00	0 ± 0.00	0.1016 ± 0.01
Bal-VAE-Attention (Ours)	0.99 ± 0.01	0.99 ± 0.01	0.99 ± 0.01	0.99 ± 0.01

4.7. Discussion

We propose a method that incorporates data augmentation into attention and proves its superiority.

Credit card fraud detection can be generally implemented through traditional machine learning methods. Nevertheless, their precision and the capabilities to discover positive samples are usually suboptimal. The most important reasons for our model's perfect precision are: (1) the newly designed oversampling method considers the imbalanced distribution of genuine and fraudulent cases, and it resolves this problem theoretically; (2) we deepen the structure of attention and mine the data from a more complex and intact perspective. It reduces the noise of original data to the largest extent.

5. Conclusion

We propose Bal-VAE-Attention. This method exceeds the other state-of-the-art credit card fraud detection baselines. We enhance and improve the detection performance by adding novel augmentation and complex attention mechanisms successfully.

In our future work, We may harness the power of generative AI tools to transform words or numbers into graphs automatically and create more novel approaches for fraud detection. For instance, the detection results might be shown in a graphic form or there might be a digital banker who can assess the fraud detection in one's mobile phone.

CRedit authorship contribution statement

Si Shi: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Wuman Luo:** Writing – review & editing, Supervision, Resources, Funding acquisition, Conceptualization. **Giovanni Pau:** Writing – review & editing, Supervision, Resources, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix. Mathematical proof

Proof. We provide mathematical proof for Theorem 3.1.

The optimization problem can be denoted as $\arg \min_{\theta} \|L1 + Regularization\|$, where θ denotes the learned parameter for the reconstruction and regularization loss terms.

Therefore, we compare the original regularization loss function $L2$ and the novel $L3$, the one with smaller amount since a certain step will be more proper.

$$n(m_i) \ll n(n_i) \Rightarrow 0 < \frac{n(m_i)}{n(n_i)} \ll 1$$

$$\Rightarrow \begin{cases} \ln(\frac{n(m_i)}{n(n_i)})\mu_{(i)}^2(x) \ll \mu_{(i)}^2(x), \\ \frac{1}{2} \ln(\frac{n(n_i)}{n(m_i)}) \ln \sigma_{(i)}^2(x) \gg \ln \sigma_{(i)}^2(x). \end{cases}$$

Let $\sigma_{(i)}^2(x) = y$,

$$((\frac{n(m_i)}{n(n_i)})^{\ln \sigma_{(i)}^2(x)})'_y = \ln(\frac{n(m_i)}{n(n_i)}) \cdot (\frac{n(m_i)}{n(n_i)})^{\ln y} \cdot \frac{1}{y}$$

Because $\ln(\frac{n(m_i)}{n(n_i)}) < 0$, $\frac{n(n_i)}{n(m_i)} > 0$, and $y > 0$, the derivative of the whole expression < 0 . Also, $x \sim N(\mu_{(i)}, \sigma_{(i)}^2)$, y and x have the same monotonicity. Thus, $((\frac{n(m_i)}{n(n_i)})^{\ln \sigma_{(i)}^2(x)})'_x < 0$.

Therefore, $(\frac{n(m_i)}{n(n_i)})^{\ln \sigma_{(i)}^2(x)} \ll \sigma_{(i)}^2(x)$ when x reaches a certain amount.

Above all, all the terms in $L3$ will be smaller than $L2$ from a certain training time step.

Therefore, $L3 = o(L2)$. The theorem is established. $\square$

Data availability

Data will be made available on request.

References

- [1] S. Bhattacharyya, S. Jha, K. Tharakunnel, J.C. Westland, Data mining for credit card fraud: A comparative study, *Decis. Support Syst.* 50 (3) (2011) 602–613, <http://dx.doi.org/10.1016/j.dss.2010.08.008>.
- [2] N. Report, Card fraud worldwide, *Nilson Rep.* 1209 (2021).
- [3] I. Benchaji, S. Douzi, B. El Ouahidi, J. Jaafari, Enhanced credit card fraud detection based on attention mechanism and LSTM deep model, *J. Big Data* 8 (2021) 1–21, <http://dx.doi.org/10.1186/s40537-021-00541-8>.
- [4] V. Narayan, S. Ganapathisamy, Hybrid sampling and similarity attention layer in bidirectional long short term memory in credit card fraud detection, *Int. J. Intell. Eng. Syst.* 15 (6) (2022) <http://dx.doi.org/10.22266/ijies2022.1231.04>.
- [5] D. Cheng, S. Xiang, C. Shang, Y. Zhang, F. Yang, L. Zhang, Spatio-temporal attention-based neural network for credit card fraud detection, in: *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 34, 2020, pp. 362–369, <http://dx.doi.org/10.1609/aaai.v34i01.5371>.
- [6] Y. Xie, G. Liu, C. Yan, C. Jiang, M. Zhou, Time-aware attention-based gated network for credit card fraud detection by extracting transactional behaviors, *IEEE Trans. Comput. Soc. Syst.* 10 (3) (2023) 1004–1016, <http://dx.doi.org/10.1109/tcss.2022.3158318>.
- [7] R. Cao, G. Liu, Y. Xie, C. Jiang, Two-level attention model of representation learning for fraud detection, *IEEE Trans. Comput. Soc. Syst.* 8 (6) (2021) 1291–1301, <http://dx.doi.org/10.1109/tcss.2021.3074175>.
- [8] J. Chen, Y. Shen, R. Ali, Credit card fraud detection using sparse autoencoder and generative adversarial network, in: *2018 IEEE 9th Annual Information Technology, Electronics and Mobile Communication Conference, IEMCON*, IEEE, 2018, pp. 1054–1059, <http://dx.doi.org/10.1109/iemcon.2018.8614815>.
- [9] H. Tingfei, C. Guangquan, H. Kuihua, Using variational auto encoding in credit card fraud detection, *IEEE Access* 8 (2020) 149841–149853, <http://dx.doi.org/10.1109/access.2020.3015600>.
- [10] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A.N. Gomez, L. Kaiser, I. Polosukhin, Attention is all you need, *Adv. Neural Inf. Process. Syst.* 30 (2017) doi:<https://dl.acm.org/doi/10.5555/3295222.3295349>.
- [11] I. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, Y. Bengio, Generative adversarial networks, *Commun. ACM* 63 (11) (2020) 139–144, <http://dx.doi.org/10.1145/3422622>.
- [12] N.V. Chawla, K.W. Bowyer, L.O. Hall, W.P. Kegelmeyer, SMOTE: synthetic minority over-sampling technique, *J. Artificial Intelligence Res.* 16 (2002) 321–357, <http://dx.doi.org/10.1613/jair.953>.
- [13] D.P. Kingma, M. Welling, Auto-encoding variational bayes, 2013, <http://dx.doi.org/10.48550/arXiv.1312.6114>, arXiv preprint arXiv:1312.6114.
- [14] S.A. Ebiaredoh-Mienye, E. Esenogho, T.G. Swart, Artificial neural network technique for improving prediction of credit card default: A stacked sparse autoencoder approach, *Int. J. Electr. Comput. Eng.* 11 (5) (2021) 4392, <http://dx.doi.org/10.11591/ijece.v11i5.pp4392-4402>.
- [15] A. Pandey, A. Bhatraju, S. Markam, D. Bhatt, Adversarial fraud generation for improved detection, in: *Proceedings of the Third ACM International Conference on AI in Finance*, 2022, pp. 123–129, <http://dx.doi.org/10.1145/3533271.3561723>.
- [16] S. Misra, S. Thakur, M. Ghosh, S.K. Saha, An autoencoder based model for detecting fraudulent credit card transaction, *Procedia Comput. Sci.* 167 (2020) 254–262, <http://dx.doi.org/10.1016/j.procs.2020.03.219>.
- [17] R. Cao, J. Wang, M. Mao, G. Liu, C. Jiang, Feature-wise attention based boosting ensemble method for fraud detection, *Eng. Appl. Artif. Intell.* 126 (2023) 106975, <http://dx.doi.org/10.1016/j.engappai.2023.106975>.
- [18] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, T.-Y. Liu, Lightgbm: A highly efficient gradient boosting decision tree, *Adv. Neural Inf. Process. Syst.* 30 (2017) doi:<https://dl.acm.org/doi/10.5555/3294996.3295074>.
- [19] L. Breiman, Random forests, *Mach. Learn.* 45 (2001) 5–32, <http://dx.doi.org/10.1023/A:1010933404324>.
- [20] T. Chen, C. Guestrin, Xgboost: A scalable tree boosting system, in: *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016, pp. 785–794, <http://dx.doi.org/10.1145/2939672.2939785>.
- [21] G. Pang, C. Shen, A. van den Hengel, Deep anomaly detection with deviation networks, in: *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, 2019, pp. 353–362, <http://dx.doi.org/10.1145/3292500.3330871>.
- [22] Z. Meng, Y. Xie, J. Sun, Detecting credit card fraud by generative adversarial networks and multi-head attention neural networks, *IAENG Int. J. Comput. Sci.* 50 (2) (2023) 42–59.
- [23] S. Jiang, R. Dong, J. Wang, M. Xia, Credit card fraud detection based on unsupervised attentional anomaly detection network, *Systems* 11 (6) (2023) 305, <http://dx.doi.org/10.3390/systems11060305>.
- [24] T. Fawcett, An introduction to ROC analysis, *Pattern Recognit. Lett.* 27 (8) (2006) 861–874, <http://dx.doi.org/10.1016/j.patrec.2005.10.010>.